The empirical recurrence for OEIS A278154, recovered from its tail

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Abstract

OEIS A278154 records "Empirical recurrence of order 89" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 104$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{89} = 20 \neq 0$, so no further tail satisfies anything shorter; any order-89 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A278154 is "Number of nX5 0..1 arrays with every element both equal and not equal to some elements at offset (-1,-1) (-1,0) (-1,1) (0,-1) (0,1) (1,-1) or (1,0), with upper left element zero.". It has offset 1 and begins

$$0, 75, 1759, 43876, 1118946, 28772962, 736017223, 18830978857, \ \ldots$$

The entry states:

> Empirical recurrence of order 89 (see link above)

As of the "Last modified" line on the live entry (Nov 13 2016 06:35:58, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 104$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 104$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 89: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 89.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 208$ exact terms from $n = 3$ on, it returns order 89 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{89} c_i \, a(n-i),$$

c_{1}	19	c_{24}	−367592266362	c_{47}	216461960584993	c_{70}	−74944767625
c_{2}	156	c_{25}	−306699401539	c_{48}	105387050910024	c_{71}	43256971496
c_{3}	333	c_{26}	−5358260796321	c_{49}	−372148176775718	c_{72}	426305160
c_{4}	778	c_{27}	13944072496560	c_{50}	337389853942767	c_{73}	−12147931178
c_{5}	−21494	c_{28}	−1557235369664	c_{51}	−147915186459999	c_{74}	5218308902
c_{6}	−138043	c_{29}	13107773334155	c_{52}	−47341338369472	c_{75}	−2742450298
c_{7}	136429	c_{30}	57075364764897	c_{53}	172904485875435	c_{76}	174910242
c_{8}	1502469	c_{31}	−67241151135250	c_{54}	−141747453825701	c_{77}	477281127
c_{9}	201983	c_{32}	99690581293902	c_{55}	69481565277692	c_{78}	−163704956
c_{10}	5757746	c_{33}	−102651947883219	c_{56}	17240598290679	c_{79}	86031170
c_{11}	23858081	c_{34}	−152972542188069	c_{57}	−54361790197133	c_{80}	−15530009
c_{12}	−90305821	c_{35}	164878696221414	c_{58}	40758101118007	c_{81}	−5582721
c_{13}	−163472442	c_{36}	−574967075423372	c_{59}	−21891267291413	c_{82}	1604
c_{14}	−436361144	c_{37}	387113677251256	c_{60}	−3925528494312	c_{83}	−169374
c_{15}	−1686090001	c_{38}	−309933168353220	c_{61}	11049678304175	c_{84}	74430
c_{16}	2793841161	c_{39}	−112841619766103	c_{62}	−7214852598780	c_{85}	55083
c_{17}	4841645718	c_{40}	635910701163476	c_{63}	4393294699398	c_{86}	−2547
c_{18}	7956845869	c_{41}	−679157346394152	c_{64}	508338572772	c_{87}	−1354
c_{19}	76369933981	c_{42}	723866801631288	c_{65}	−1578965385975	c_{88}	−147
c_{20}	−9466866312	c_{43}	−145085863538347	c_{66}	813154862538	c_{89}	20
c_{21}	−63794838452	c_{44}	−293432825149009	c_{67}	−553234243791		
c_{22}	126873996066	c_{45}	598203776802228	c_{68}	−28326869642		
c_{23}	−1469307626291	c_{46}	−605644970681725	c_{69}	176092649816		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A278154 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{89} - c_1 x^{89-1} - \cdots - c_{89}$. Its constant term is $-c_{89} = -20$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 89 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 89, so $\deg q = 89$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 89, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 89, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 20.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-89 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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